Paper 1 Analysis Table

Category	Details
Research Paper Title	*Deep Learning for Automated Breast Cancer Detection in Ultrasound: A Comparative Study of Four CNN Architectures*
Objective	To compare four CNN architectures (ResNet50, DenseNet169, InceptionV3, InceptionV4) for automatic classification of breast cancer using ultrasound images and identify the best-performing model.
Problem Statement	Ultrasound imaging is widely used for breast cancer detection but suffers from **operator dependency, variability, and low accuracy in manual diagnosis**. There is a need for **automated, accurate, and reliable AI-based diagnostic systems**.
Model Used	- ResNet50 - DenseNet169 - InceptionV3 (best) - InceptionV4
Programming / Tools	Not explicitly mentioned, but based on methodology: ➤ Likely Python ➤ Deep Learning frameworks (TensorFlow / PyTorch) ➤ Transfer Learning with ImageNet weights
Dataset	Breast Ultrasound Images Dataset (BUSI - Kaggle) - Total: 780 images - Classes: Benign (437), Malignant (210), Normal (133)
Pros	✔ High accuracy (InceptionV3 = 96.67%) ✔ Comparative analysis of multiple CNN models ✔ Uses real ultrasound data ✔ Transfer learning improves performance ✔ Shows trade-off between accuracy & computational cost
Cons	✘ Small dataset (only 780 images) ✘ No real clinical validation ✘ No explainability (XAI not implemented) ✘ Only classification (no segmentation or localization)
Challenges	⚠ Limited dataset size ⚠ Noise and low contrast in ultrasound images ⚠ Class imbalance ⚠ High computational cost for complex models ⚠ Overfitting risk

Conclusion	InceptionV3 achieved the best performance and is suitable for breast cancer classification in ultrasound images. DenseNet169 is a good lightweight alternative. AI models show strong potential for early diagnosis.
Critical Comment	This paper clearly shows that **deep learning works well for ultrasound-based cancer detection**, but it lacks **explainability**, which is critical in medical AI. For your FYP (*UltraXAI*), this is a major gap you can improve by adding **Explainable AI (like Grad-CAM)** to make decisions interpretable for doctors. Also, the dataset is small, so your project can improve reliability using **larger or multi-dataset training**.